

Job Portal Analysis – Project Report

1. Project Overview

The **Job Portal Analysis** project focuses on understanding the current Indian job market using job postings collected from major recruitment platforms such as [Naukri.com](#) and [TimesJobs](#). The project involved collecting raw job data, performing extensive data cleaning and preprocessing, conducting exploratory data analysis (EDA), and generating business insights through visualizations and dashboards.

The final dataset contained approximately **149K cleaned job records**, covering multiple industries, experience levels, salary ranges, skills, locations, and hiring companies.

2. Objective of the Project

The main objectives of this project were:

- Analyze hiring trends across industries in India.
 - Understand salary distribution across experience levels and industries.
 - Identify the most in-demand skills in the current job market.
 - Study job distribution by experience level.
 - Identify top hiring companies and locations.
 - Extract structured insights from unstructured job descriptions.
 - Build meaningful visualizations and dashboards for better decision-making.
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3. Data Collection

The dataset was created using web scraping techniques from job portals such as:

- [Naukri.com](#)
- [TimesJobs](#)

Data Fields Collected

The following information was extracted from job postings:

- Job Title
- Company Name
- Salary
- Experience Required
- Skills / Tags
- Industry

- Location
- Employment Type
- Qualification
- Job Description
- Posting Time
- Job URL

Tools & Libraries Used

- Python
 - Pandas
 - NumPy
 - BeautifulSoup
 - Requests
 - Regex (re)
 - Tableau
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4. Actions Performed

4.1 Data Cleaning

Several preprocessing steps were performed to improve data quality and consistency.

Cleaning Activities

- Removed duplicate job postings.
 - Handled missing/null values.
 - Standardized inconsistent text fields.
 - Cleaned salary and experience formats.
 - Merged datasets from multiple job portals.
 - Removed irrelevant or noisy records.
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4.2 Data Preprocessing

Experience Extraction & Binning

Experience values in different formats were standardized into numerical ranges and grouped into categories such as Entry Level, Mid Level, Senior Level, and Expert Level.

Salary Standardization

Salary values from different formats were cleaned and converted into a standardized numeric format for accurate analysis.

Industry Binning

Industries with many unique values were grouped into broader categories like IT & Software, Finance, Manufacturing, Sales, HR, and Consulting.

Feature Engineering

Important information was extracted from job descriptions using rule-based text processing:

- Degrees & Qualifications
- Soft Skills
- Responsibilities
- Must Have

This transformed unstructured text into structured analytical features.

5. Exploratory Data Analysis (EDA)

EDA was performed using Tableau dashboards and visualizations to identify trends and patterns.

Visualizations Created

- Distribution of jobs by experience level
- Salary distribution across industries
- Average salary by role
- Top hiring companies
- Top hiring locations
- Most in-demand skills
- Skill demand by experience level
- Experience vs Salary relationship

Dashboard Link:

https://public.tableau.com/app/profile/khushi.yadav3650/viz/NaukriJobPostingDashboard_updated/SkillJobProfilesInsights

6. Key Insights

6.1 Experience Level Distribution

- Senior-level jobs dominate the market with approximately **38%** of total postings.
 - Entry-level opportunities are significantly lower (~8.5%), making it difficult for freshers to enter the industry.
 - Expert and leadership roles receive the highest compensation packages.
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6.2 Salary Insights

- Manufacturing industries showed the highest average salaries.
 - Finance and IT sectors also offered competitive salaries.
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6.3 Industry Hiring Trends

- Software Development and Engineering had the highest number of job openings.
 - Sales roles remained highly active despite the growth of e-commerce.
 - Media and content-related jobs are growing steadily.
 - Manufacturing offers high salaries but fewer openings.
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6.4 Skills in Demand

The most demanded skills were:

- Programming
- Databases
- Sales
- Management

Programming and database-related roles dominated across all experience levels, highlighting the growing demand for IT and data-related careers.

6.5 Experience vs Salary

- Salary generally increases with experience.
 - Salary growth is not perfectly linear.
 - High salary outliers exist across multiple experience groups.
 - Industry and skills influence salary more than experience alone.
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6.6 Top Hiring Locations

The leading hiring cities were Bengaluru, Hyderabad, Pune and Mumbai .

Key observations:

- Bengaluru emerged as the largest hiring hub.
 - Hyderabad and Pune showed strong IT hiring activity.
 - Remote jobs still remain relevant in the market.
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6.7 Top Hiring Companies

Major hiring companies included Capgemini, NTT DATA, DBS Bank and Pfizer.

Capgemini recorded the highest hiring activity among analyzed companies.

7. Recommendations

For Job Seekers

- Focus on in-demand technical skills.
 - Regularly update GitHub and LinkedIn profiles.
 - Network actively with recruiters and professionals.
 - Ask for salary clarity during interviews.
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For Companies

- Provide transparent salary ranges.
 - Simplify application processes.
 - Clearly define job requirements.
 - Avoid vague or misleading job postings.
 - Improve hiring communication and candidate experience.
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8. Conclusion

This project successfully analyzed large-scale job market data to identify important hiring and salary trends in India. Through data cleaning, preprocessing, feature engineering, and visualization, the project transformed raw job postings into meaningful business insights.

The findings highlight:

- Strong demand for technology-related roles.
- Growing importance of programming and data skills.
- High competition for entry-level jobs.
- Significant influence of industry and skills on salary.